

# PHYSICAL SCIENCES

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## PHYSICAL SCIENCES GRADE 12 TERM 1 CONTROLLED ASSESSMENT PACK

|                 |                                                                 |
|-----------------|-----------------------------------------------------------------|
| SUBJECT         | PHYSICAL SCIENCES                                               |
| GRADE           | 12                                                              |
| ASSESSMENT TASK | PHYSICAL SCIENCES GRADE 12 TERM 1<br>CONTROLLED ASSESSMENT PACK |
| MARKS           | 50                                                              |
| TIME            | 1 hour                                                          |

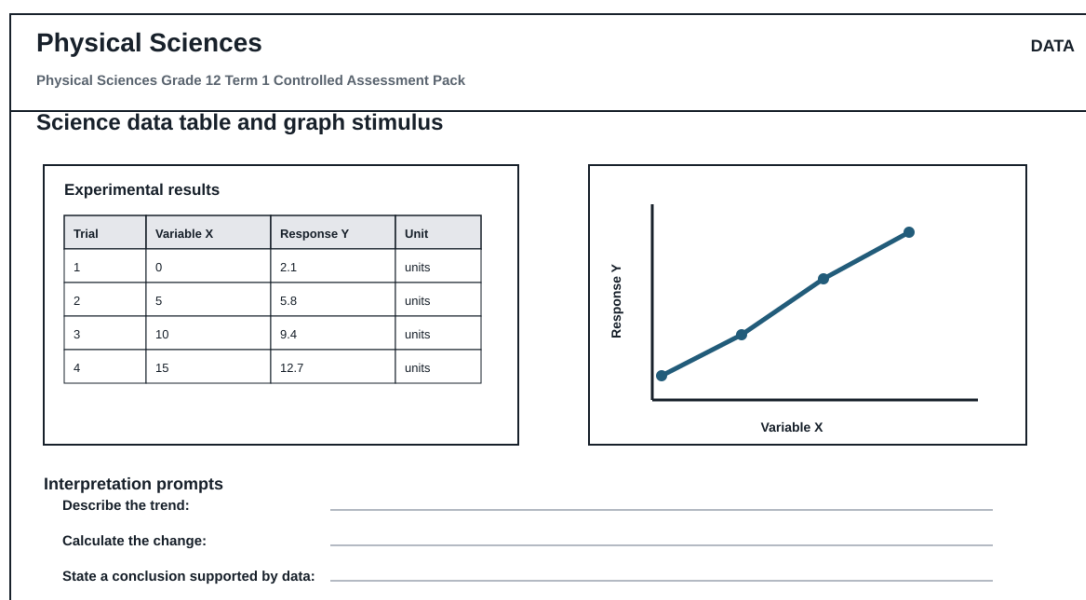
Educator approval required before classroom use.

## INSTRUCTIONS AND INFORMATION

1. Read all instructions carefully.
2. Answer all questions or complete all activities as instructed.
3. Show all working where calculations, planning, diagrams or explanations are required.
4. Use the space provided by the educator, answer booklet or learner task book.
5. Write neatly and clearly.

## VISUAL MATERIAL

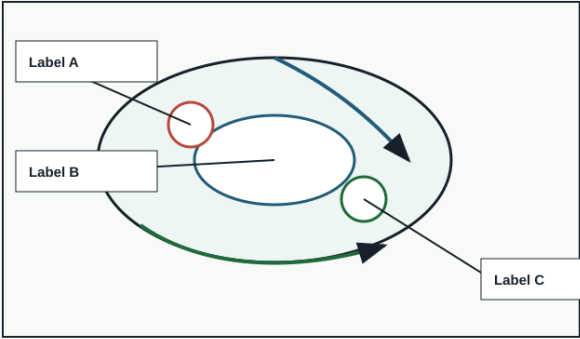
**FIGURE 1: Science Data Table And Graph Stimulus**



**FIGURE 2: Investigation Data And Method Sheet**

| Physical Sciences                                            |             | INVESTIGATION |                   |
|--------------------------------------------------------------|-------------|---------------|-------------------|
| Physical Sciences Grade 12 Term 1 Controlled Assessment Pack |             |               |                   |
| <b>Investigation method, data and conclusion</b>             |             |               |                   |
| Aim                                                          | Variables   | Method        | Results           |
| —                                                            |             |               |                   |
| Conclusion                                                   |             |               |                   |
| Reading                                                      | Observation | Unit          | Pattern / anomaly |
|                                                              |             |               |                   |
|                                                              |             |               |                   |
|                                                              |             |               |                   |
|                                                              |             |               |                   |
| Conclusion supported by data:                                |             |               |                   |

**FIGURE 3: Physical Sciences Labelled Diagram Stimulus**

| Physical Sciences                                                                              |  | DIAGRAM                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |  |
|------------------------------------------------------------------------------------------------|--|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|
| Physical Sciences Grade 12 Term 1 Controlled Assessment Pack                                   |  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |  |
| <b>Labelled science diagram stimulus</b>                                                       |  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |  |
|             |  | <p><b>Assessment actions</b></p> <p>Physical Sciences Grade 12 Term 1<br/>Controlled Assessment Pack</p> <div style="margin-top: 10px;"> <div style="border: 1px solid black; padding: 2px 5px; margin-bottom: 5px;">Identify labelled structures</div> <div style="border: 1px solid black; padding: 2px 5px; margin-bottom: 5px;">Explain the process</div> <div style="border: 1px solid black; padding: 2px 5px; margin-bottom: 5px;">Relate structure to function</div> <div style="border: 1px solid black; padding: 2px 5px;">Predict effect of a change</div> </div> |  |
| Use this as the supplied diagram; replace labels with topic-specific structures during review. |  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |  |

## QUESTIONS

### QUESTION 1: SECTION A: CONCEPTS AND DATA

**Instructions:** Use the supplied data, diagram or graph.

**Stimulus:** Data/diagram stimulus linked to mechanics, momentum, vertical projectile motion and Newton's laws.

**1.1.** Define one key concept from the term focus. **(2)**

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**1.2.** Describe the trend or relationship shown by the supplied data. **(5)**

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**1.3.** Explain the process or principle using correct terminology. **(8)**

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**1.4.** Calculate or infer a result from the supplied values and show working where needed. **(5)**

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## **QUESTION 2: SECTION B: INVESTIGATION DESIGN**

**Instructions:** Critique method, variables and evidence.

**Stimulus:** Investigation focus: formula selection, substitution, units and sign convention.

**2.1.** State a suitable aim or hypothesis for the investigation. **(3)**

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**2.2.** Identify independent, dependent and controlled variables. **(6)**

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**2.3.** Evaluate validity and reliability and suggest one improvement. **(6)**

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### QUESTION 3: SECTION C: APPLICATION

**Instructions:** Use subject knowledge and supplied evidence.

**Stimulus:** Application focus: investigation or data-response task using motion evidence.

**3.1.** Answer an integrated explanation question using evidence from the stimulus. **(15)**

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## MARKING GUIDELINE

### TASK 1: SECTION A: CONCEPTS AND DATA

**1.1.** Define one key concept from the term focus. **(2)**

- Definition scientifically accurate.

**1.2.** Describe the trend or relationship shown by the supplied data. **(5)**

- Trend described.
- Evidence cited.

**1.3.** Explain the process or principle using correct terminology. **(8)**

- Correct terms.
- Logical explanation.

**1.4.** Calculate or infer a result from the supplied values and show working where needed. **(5)**

- Working shown.
- Answer accurate.

### TASK 2: SECTION B: INVESTIGATION DESIGN

**2.1.** State a suitable aim or hypothesis for the investigation. **(3)**

- Aim/hypothesis testable.

**2.2.** Identify independent, dependent and controlled variables. **(6)**

- Variables correctly classified.

**2.3.** Evaluate validity and reliability and suggest one improvement. **(6)**

- Validity/reliability explained.
- Improvement practical.

### TASK 3: SECTION C: APPLICATION

**3.1.** Answer an integrated explanation question using evidence from the stimulus. **(15)**

- Evidence used.
- Explanation coherent.
- Conclusion supported.

## RUBRIC

| Criterion                        | Marks | Descriptor                              |
|----------------------------------|-------|-----------------------------------------|
| Subject accuracy and terminology | 12    | Uses accurate concepts and terminology. |
| Evidence, data or source         | 12    | Uses supplied evidence                  |

|                              |    |                                             |
|------------------------------|----|---------------------------------------------|
| use                          |    | appropriately.                              |
| Reasoning and explanation    | 12 | Shows logical, grade-appropriate reasoning. |
| Communication and completion | 14 | Answers are clear, complete and structured. |

## EDUCATOR REVIEW CHECKLIST

- Confirm CAPS term and topic fit before classroom use.
- Insert or approve final source material where the pack requests educator-supplied sources.
- Check mark allocation, memo wording and learner level.
- Confirm visuals are suitable and not treated as official source material unless sourced separately.